

Experiment No. - 08

AIM:- To study about the configuration of link state OSPF routing protocol by using Cisco Packet Tracer.

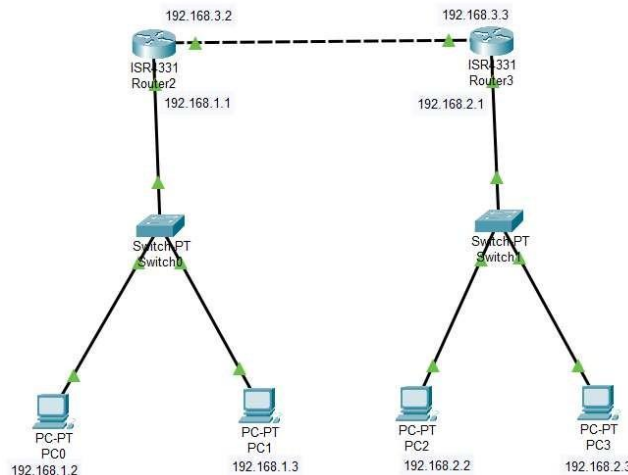
● STEPS :

1. Set Up the Network Topology:

- Place the routers (ISR331, Router2, Router3), switches (SwitchPT, Switch), and PCs (PC0, PC1, PC2, PC3) on the workspace.
- Connect the devices using appropriate cables (e.g., Copper Straight-Through for connecting PCs to switches, and Serial DCE for connecting routers).

2. Assign IP Addresses:

- Assign IP addresses to all interfaces as per the details in your file. For example:
 - Router2: 192.168.1.1 and 192.168.3.2
 - Router3: 192.168.2.1 and 192.168.3.3
 - PCs: Assign IPs like 192.168.1.2, 192.168.1.3, 192.168.2.2, 192.168.2.3 with appropriate subnet masks.



3. Access Router Configuration:

- Click on each router to open its configuration window.
- Go to the "CLI" tab to access the command-line interface.

4. Enable OSPF on Routers:

- Enter global configuration mode by typing `enable` and then `configure terminal`.
- Enable OSPF on each router. For example, on Router2:

```
router ospf 1
```

```
network 192.168.1.0 0.0.0.255 area 0
```

```
network 192.168.3.0 0.0.0.255 area 0
```

- Repeat similar commands on Router3:

```
router ospf 1
```

```
network 192.168.2.0 0.0.0.255 area 0
```

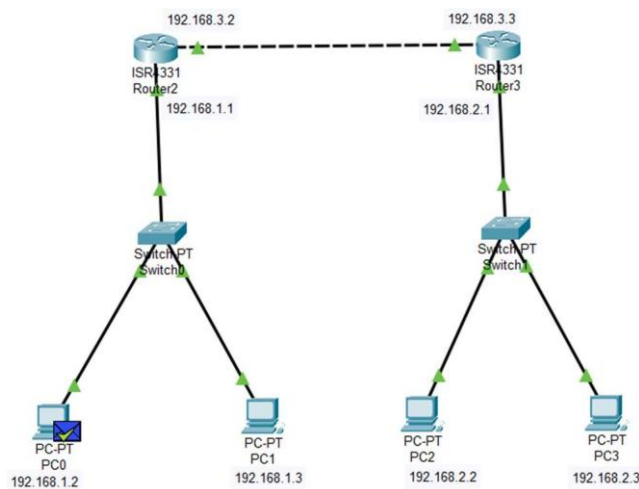
```
network 192.168.3.0 0.0.0.255 area 0
```

5. Verify OSPF Configuration:

- After configuring OSPF on all routers, verify the configuration by typing `show ip ospf neighbor` to see if the routers have formed adjacencies.
- Use `show ip router` to check the routing table and ensure that OSPF routes are being propagated.

6. Test Connectivity:

- Use the `ping` command from one PC to another to test connectivity across the network. For example, from PC0 (192.168.1.2) to PC2 (192.168.2.2).



7. Save Configuration:

- Save the configuration on each router by typing `write memory` or `copy running-config startup-config`.